

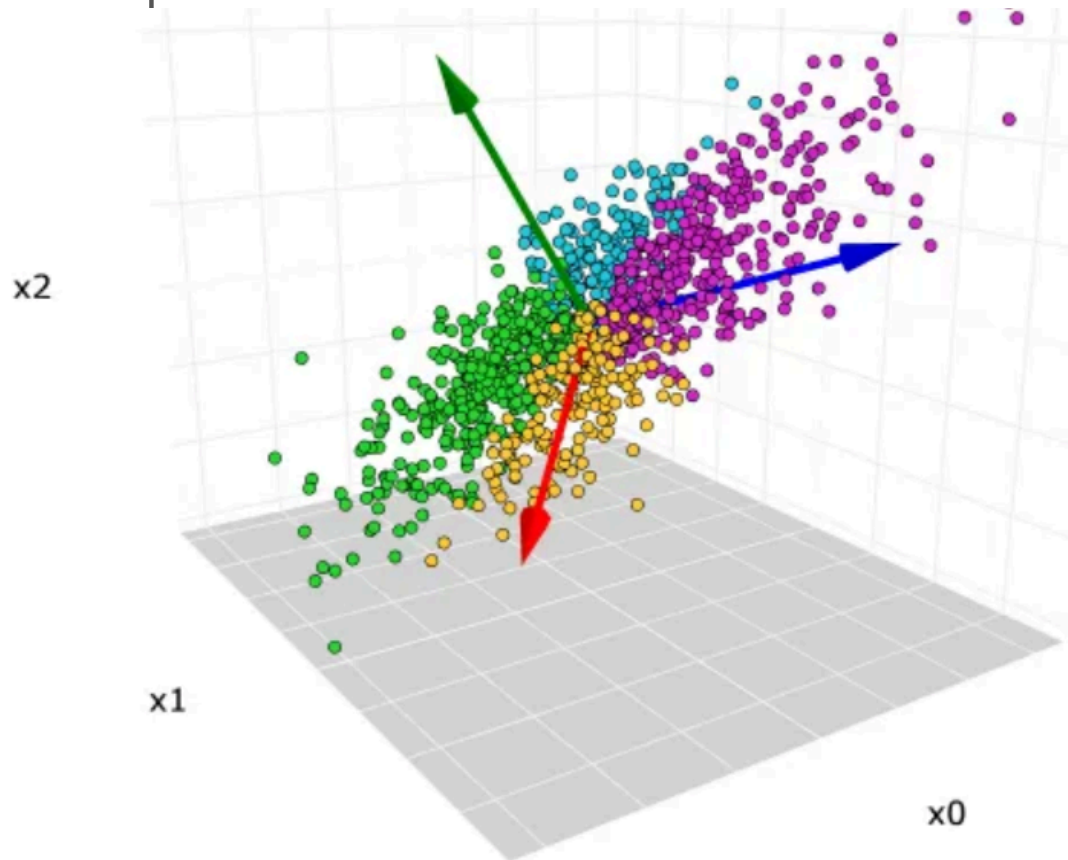
Multivariate statistics 1: Principal Component Analysis (PCA)

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Objectives of lecture

- Understand and implement Principal Component Analysis (PCA): a technique that transforms high-dimensional data into lower-dimensions while retaining as much information as possible.
- Example. Can reduce from 3 to 2 dimensions:



The principal components (PC.s)

Contents

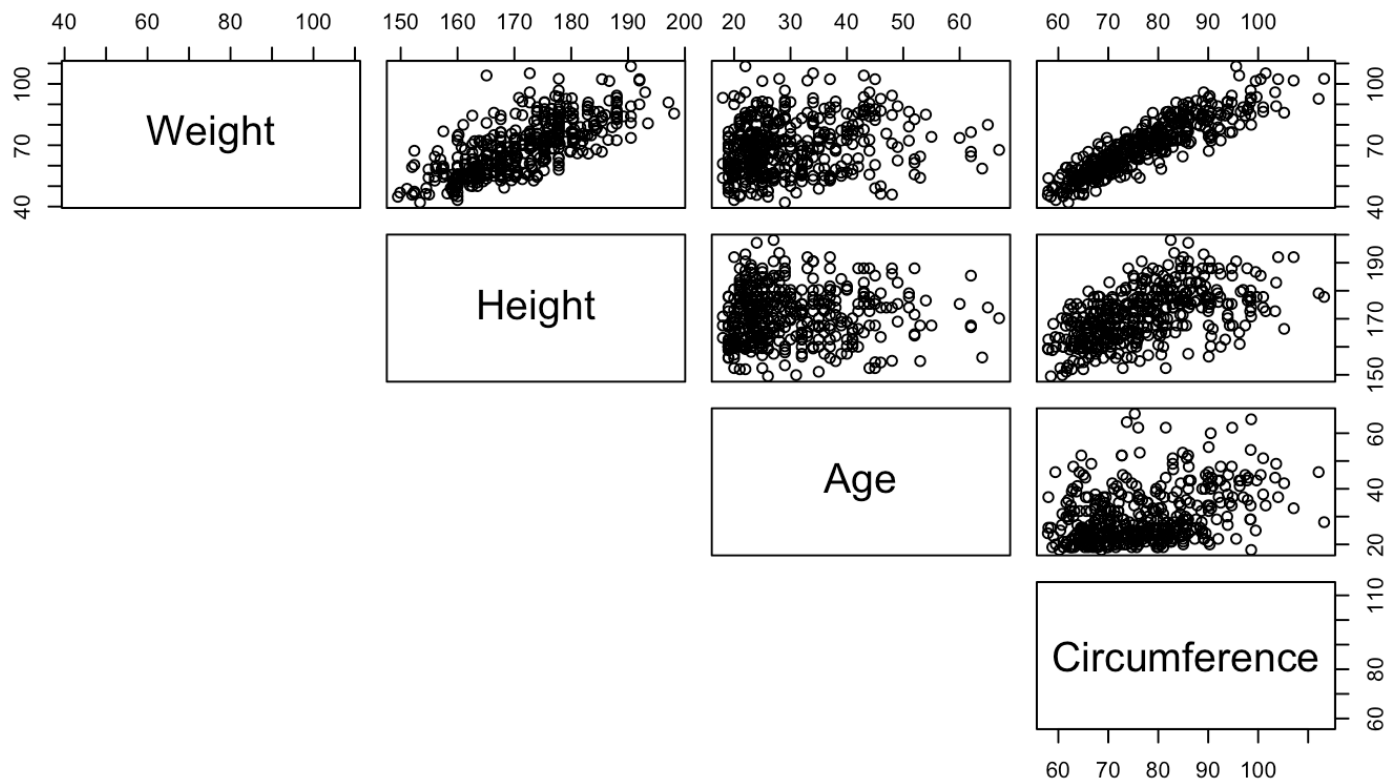
- Linear combinations
- Principal components (PC's)
- Making sense of high dimensional data

Linear Combinations

- Weighted sums of variables
- A new variable representing most of the information of others
- A very useful tool in the analysis of high-dimensional data
 - Exploration of data, generating hypotheses
 - Regression analysis when $p > n$, i.e., more variables than cases
 - Regression analysis when predictor variables are collinear (correlation close to 1)

The bodydata example

```
pairs(bodydata, lower.panel = NULL)
```



A new “HeCirc” variable for predicting weight

- The variables Height and Circumference are correlated.
- They tell much of the same about a persons weight.
- Redundancy!?
- Let’s make a new variable called “HeCirc” as the average of these:

$$HeCirc = 0.5 * Height + 0.5 * Circumference$$

Height, Circumference and average (HeCirc)

```
bodydata$HeCirc <- 0.5 * bodydata$Height +  
                  0.5 * bodydata$Circumference  
head(bodydata[,c("Height", "Circumference", "HeCirc")])
```

```
##   Height Circumference HeCirc  
## 1  174.0           71.5 122.75  
## 3  193.5           83.2 138.35  
## 4  186.5           77.8 132.15  
## 5  187.2           80.0 133.60  
## 6  181.5           82.5 132.00  
## 7  184.0           82.0 133.00
```


Centered variables, average subtracted

```
# center = TRUE: subtracts average
# scale = TRUE: gives variance = 1
body2 <- as.data.frame(scale(bodydata, center = TRUE,
                             scale = FALSE))
head(body2[,c("Height", "Circumference", "HeCirc")])
```

```
##      Height Circumference   HeCirc
## 1  2.692138   -5.4083538 -1.358108
## 3 22.192138    6.2916462 14.241892
## 4 15.192138    0.8916462  8.041892
## 5 15.892138    3.0916462  9.491892
## 6 10.192138    5.5916462  7.891892
## 7 12.692138    5.0916462  8.891892
```

About linear combinations: General case

- Assume there are k random variables, $Z_1, Z_2, \dots, Z_k$ and that these have been centered by subtracting the mean. Ex: $X_1 = (Z_1 - \bar{Z}_1)$.
- A standardized linear combination, $PC1$, is given by

$$PC1 = w_{11}X_1 + w_{12}X_2 + \dots + w_{1k}X_k$$

where $\sum_j w_{1j}^2 = 1$. The w_{ij} 's are often called weights or loadings.

- In Principal Component Analysis we seek the $PC1$ with the largest variance .

The first principal component PC1

X_1 = Height, X_2 = Circumference. The first principal component:

$$PC1 = -0.606X_1 - 0.795X_2$$

- Note that $(-0.606)^2 + (-0.795)^2 = 1$
- This is the optimal combination of Height and Circumference and better than just taking the average. It's optimal because the variance

$$\text{var}(PC1) = \dots = 164.6$$

is maximal; all other linear combinations have smaller variance, i.e., explain less.

Further components

- In PCA we seek further components which maximize variance but being orthogonal (at a right angle) to the previous components.
- The second component PC2 is

$$PC2 = w_{21}X_1 + w_{22}X_2 + \cdots + w_{2k}X_k$$

where $\sum_j w_{2j}^2 = 1$ AND $\sum_j w_{1j} \cdot w_{2j} = 0$.

It is given by:

$$PC2 = -0.795X_1 + 0.606X_2$$

Variance

$$\text{var}(PC2) = 46.6$$

PCA in R

```
(pca1 <- prcomp(~ Height + Circumference, data = bodydata))
```

```
## Standard deviations (1, ..., p=2):
```

```
## [1] 12.828601  6.825743
```

```
##
```

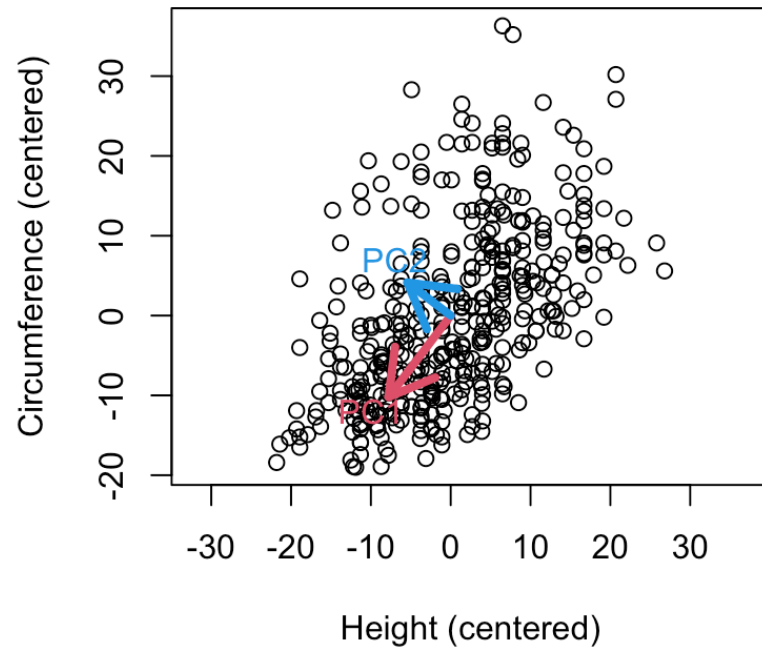
```
## Rotation (n x k) = (2 x 2):
```

```
##                PC1          PC2
```

```
## Height          -0.6063887 -0.7951684
```

```
## Circumference -0.7951684  0.6063887
```

Geometrical interpretation



Geometrical interpretation continued

- The longer red arrow of length $SD(PC1)$ points in the direction of the first principal component.
- The shorter orthogonal blue arrow of length $SD(PC2)$ points in the direction of the second principal component.

Mathematics I. PCA is by default based on the covariance matrix

```
(covMat = cov(bodydata[,c("Height", "Circumference")]))
```

```
##              Height Circumference
## Height          89.97368         56.8888
## Circumference  56.88880         121.1901
```


Mathematics II. Eigenvalues: variances.

Eigenvectors: principal components

```
eigen(covMat) # direction of eigenvectors arbitrary
```

```
## eigen() decomposition
## $values
## [1] 164.57299  46.59077
##
## $vectors
##           [,1]      [,2]
## [1,] 0.6063887 -0.7951684
## [2,] 0.7951684  0.6063887
```

- The eigenvalues of the covariance matrix are the variances of the principal components.
- The eigenvectors are the the principal components.

Amount of explained variation

Each component “explains” an independent part of the original variability

```
summary(pca1)
```

```
## Importance of components:
```

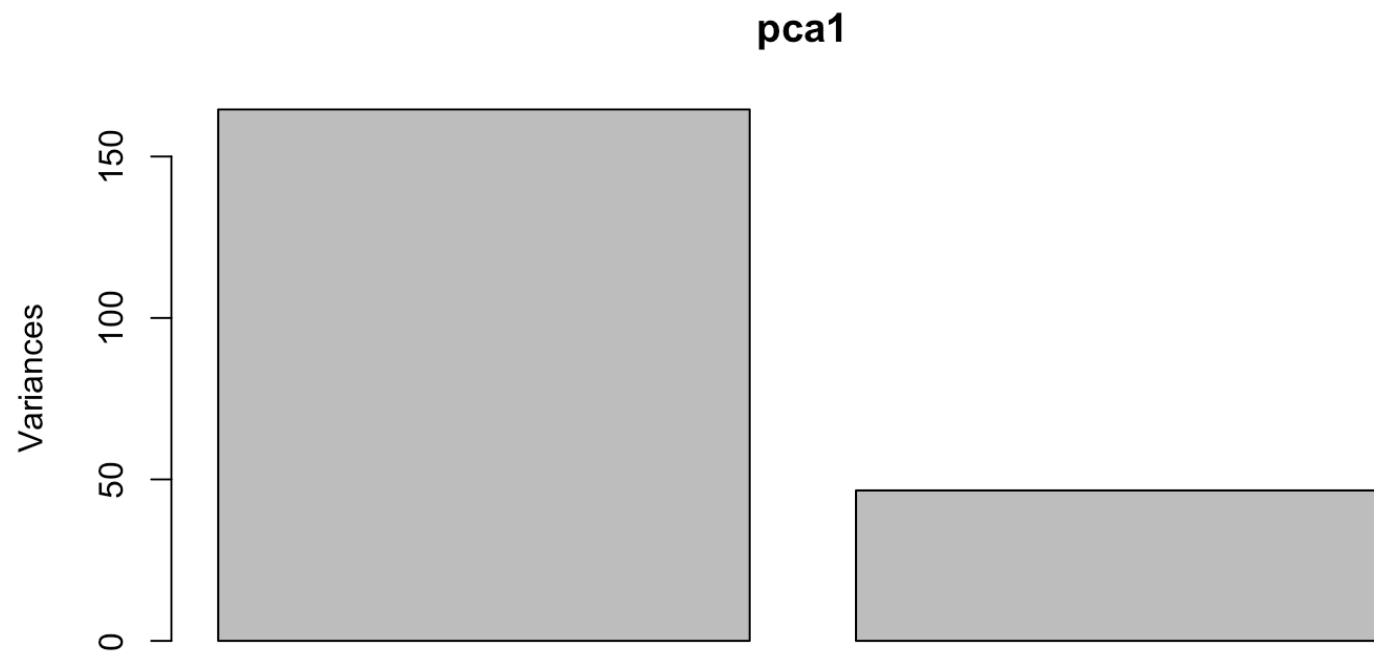
```
##              PC1      PC2
## Standard deviation    12.8286 6.8257
## Proportion of Variance 0.7794 0.2206
## Cumulative Proportion 0.7794 1.0000
```

Computation: $12.83^2 / (12.83^2 + 6.83^2) = 0.78$.

- Important information when we decide on how many principal components to keep, i.e., how much the dimension can be reduced. This can be plotted:

The scree plot

`plot(pca1)`



The scores

The observed values for the PC's for each observation are called scores

```
## Height.Centered Circumference.Centered PC1 PC2
## 1 2.692 -5.408 2.668 -5.42
```

Check. Scores on PC1 and PC2 for first observation for covariance based PCA on centered data:

$$-0.606 * 2.692 - 0.795 * (-5.408)$$

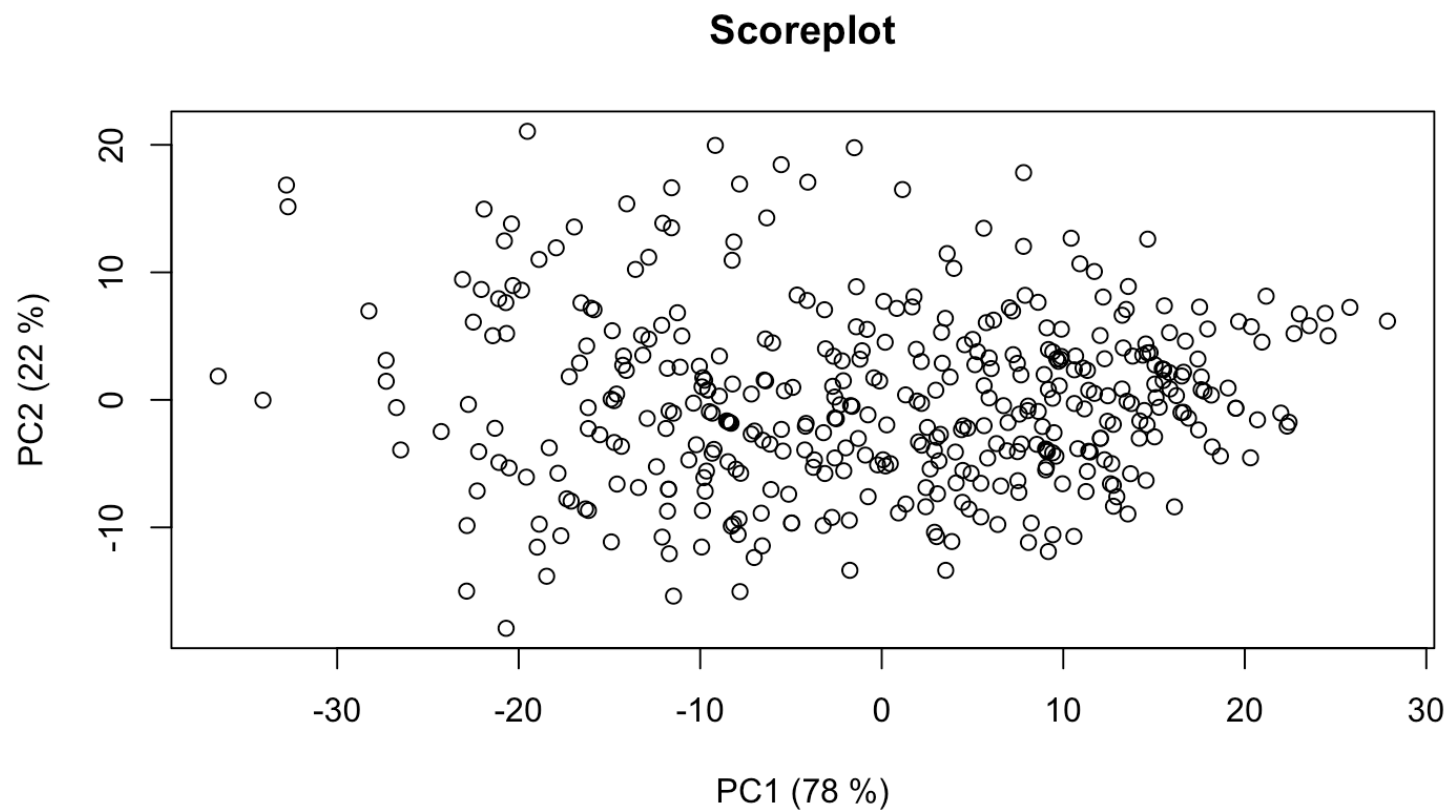
```
## [1] 2.668008
```

$$-0.795 * 2.692 + 0.606 * (-5.408)$$

```
## [1] -5.417388
```

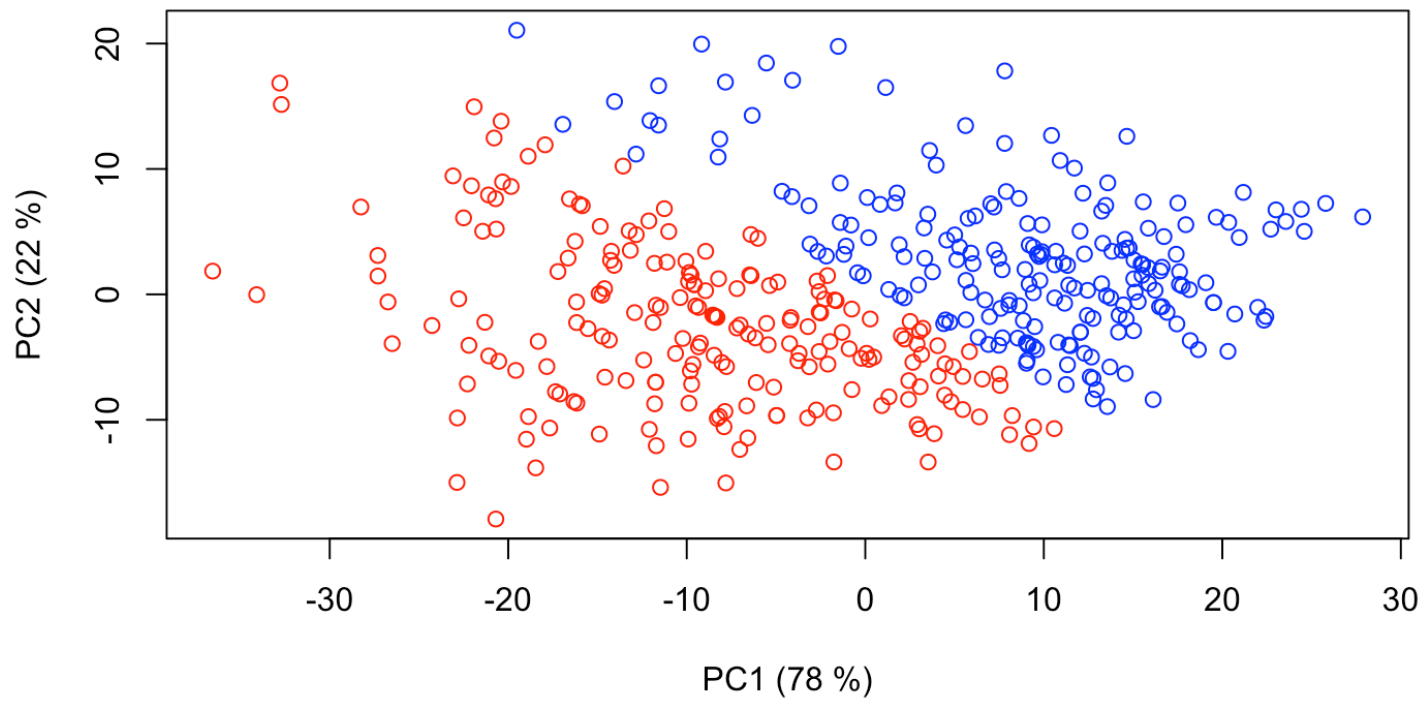
The score plot : observations

```
scoreplot(pca1, main = 'Scoreplot', comps = c(1,2))
```



Score plot. Above average tall colored red.

```
above <- ifelse(bodydata$Height > mean(bodydata$Height),  
               "red", "blue")  
scoreplot(pca1, comps = c(1,2), col = above)
```



Identify observations

```
scoreplot(pca1, main = 'Scoreplot',  
          comps = c(1,2), identify = TRUE)
```

- We can find observations by clicking and look for extremes, clusters, patterns,...

Scores for correlation based PCA

```
pca.corr <- prcomp(~ Height + Circumference,  
                  scale = TRUE, data = bodydata)
```

```
pca.corr$rotation # loadings
```

```
##              PC1      PC2  
## Height      -0.7071068 -0.7071068  
## Circumference -0.7071068  0.7071068
```

```
round(head(cbind(bodydata[,c(2,4)],pca.corr$x),1),3)
```

```
##   Height Circumference  PC1  PC2  
## 1    174          71.5 0.147 -0.548
```


How scores are calculated for correlation based PCA

- First standardise (mean = 0, SD = 1) variables for Height, Z1, and Circumference Z2

```
attach(bodydata) #Allows Height for bodydata$Height  
m1 <- mean(Height)  
sd1 <- sqrt(var(Height))  
m2 <- mean(Circumference)  
sd2 <- sqrt(var(Circumference))  
Z1 <- (Height-m1)/sd1; Z2 <- (Circumference-m2)/sd2  
PC1 = (-0.707) * Z1 + (-0.707) * Z2  
PC2 = (-0.707) * Z1 + 0.707 * Z2  
c(PC1[1], PC2[1]) #Confirms scores on previous page
```

```
## [1] 0.1466778 -0.5479961
```

```
detach(bodydata) # turns of attach
```

Example - The complete bodydata set

```
pca2 <- prcomp(~ Age + Circumference + Height + Weight,  
               data = bodydata, center = T, scale = T)  
print(pca2, digits = 3)
```

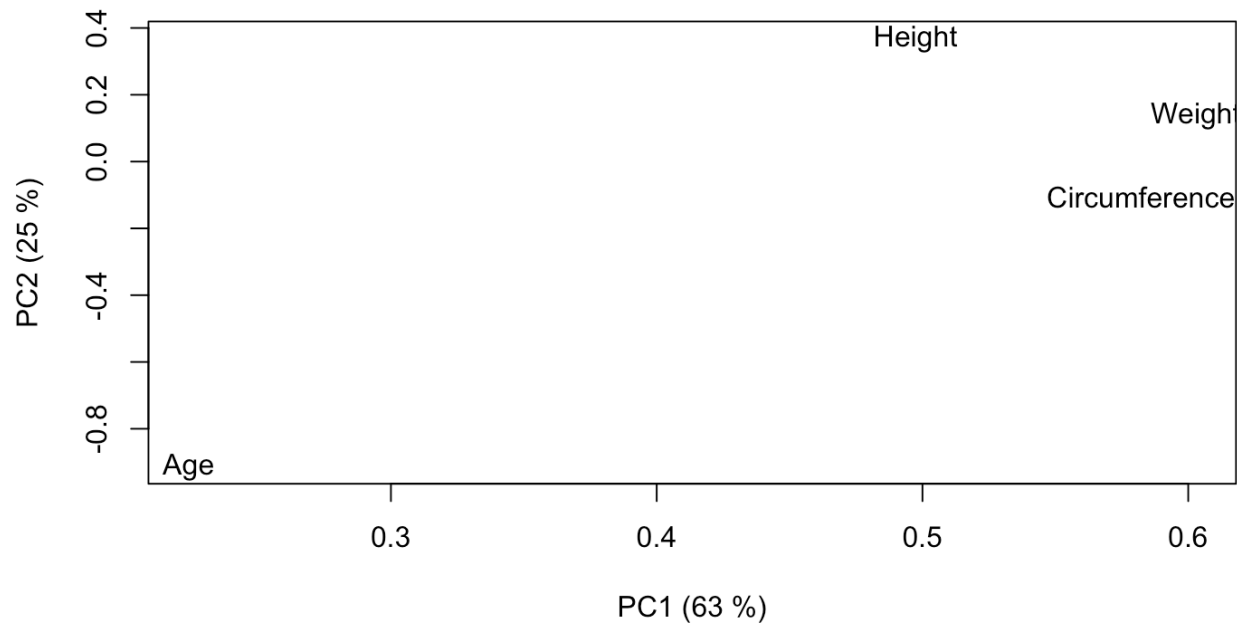
```
## Standard deviations (1, ..., p=4):  
## [1] 1.592 0.997 0.642 0.247  
##  
## Rotation (n x k) = (4 x 4):  
##  
##           PC1    PC2    PC3    PC4  
## Age         0.224 -0.913  0.325  0.0996  
## Circumference 0.582 -0.106 -0.507 -0.6266  
## Height       0.497  0.368  0.756 -0.2122  
## Weight       0.603  0.138 -0.255  0.7433
```

Importance of original variables in PCA, the loadings

- The weights $w_{11} = 0.224, \dots, w_{14} = 0.603$ are called the loadings for PC1.
- Large loadings indicate large contribution from a variable to the PC
- Variables with similar “information” (correlated) will have similar loadings.
- An overview of variable grouping and inter-dependence is quickly displayed in a loadings plot
- Plot of loadings from one component against another, e.g. for PC1 vs PC2, may be very informative.

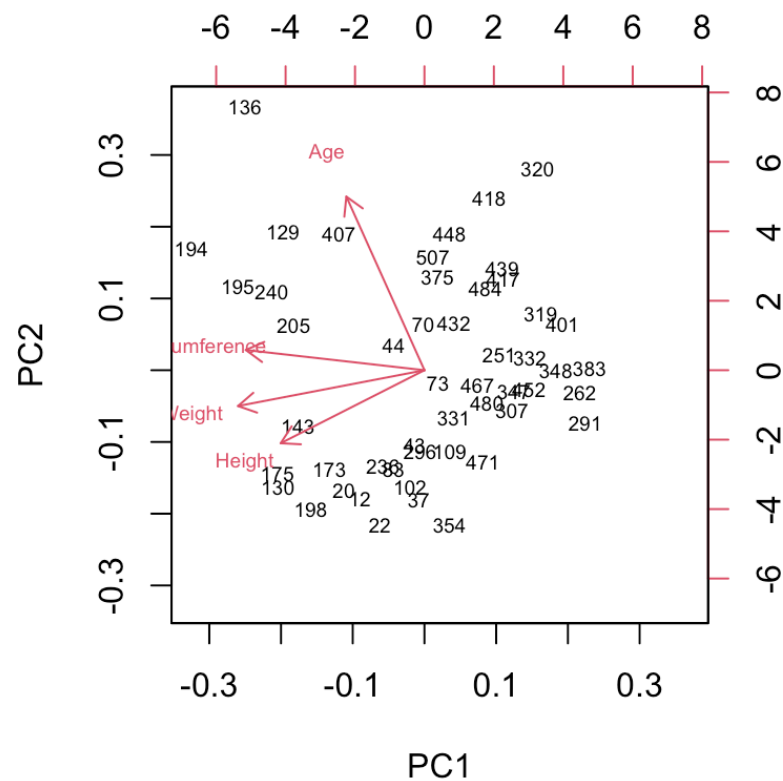
The loading plot: variables

```
loadingplot(pca2, comps = c(1,2),  
            scatter = TRUE, labels = 'names')
```



Biplot - Scores and loadings overlaid

```
foo <- prcomp(~ Age + Circumference + Height + Weight,  
  scale = T, center = T, data = bodydata[sample(1:407,50),])  
biplot(foo, choices = c(1,2), cex = 0.60)
```



Comments on plots

- Score plot: Shows observations. We look for similar observations, clusters, patterns, deviating observations.
- Loading plot: Shows variables. As above, but for variables. If a principal component is interpreted (say for ranking), we can see the influence of different variables
- Biplot: combines the above plots.
- If the PC-s we are considering, typically the first two, explain a sufficient amount of the variability, say 90%, interpretation may be promising.

Example, the Track and Field data

National records (in minutes) on several running distances for 54 nations (men).

Distances: 100m , 200m , 400m, 800m, 1500m, 5000m, 10000m and marathon.

```
head(TrackRecords,5)
```

##		X100m	X200m	X400m	X800m	X1500m	X5000m	X10000m	Marathon
##	Argentina	0.170	0.340	0.770	1.77	3.68	13.33	27.65	129.57
##	Australia	0.166	0.334	0.740	1.74	3.53	12.93	27.53	127.51
##	Austria	0.169	0.341	0.763	1.77	3.58	13.26	27.72	132.22
##	Belgium	0.169	0.336	0.750	1.73	3.57	12.83	26.87	127.20
##	Bermuda	0.171	0.338	0.754	1.79	3.70	14.64	30.49	146.37

PCA - standardized variables

- One second is different for 100m and marathon:
 - When scales differ, we standardise variables so that variances are 1.
 - Equivalently, we run PCA on the correlation matrix rather than the covariance matrix

```
pca3 <- prcomp(~. , scale. = TRUE, data = TrackRecords)
```

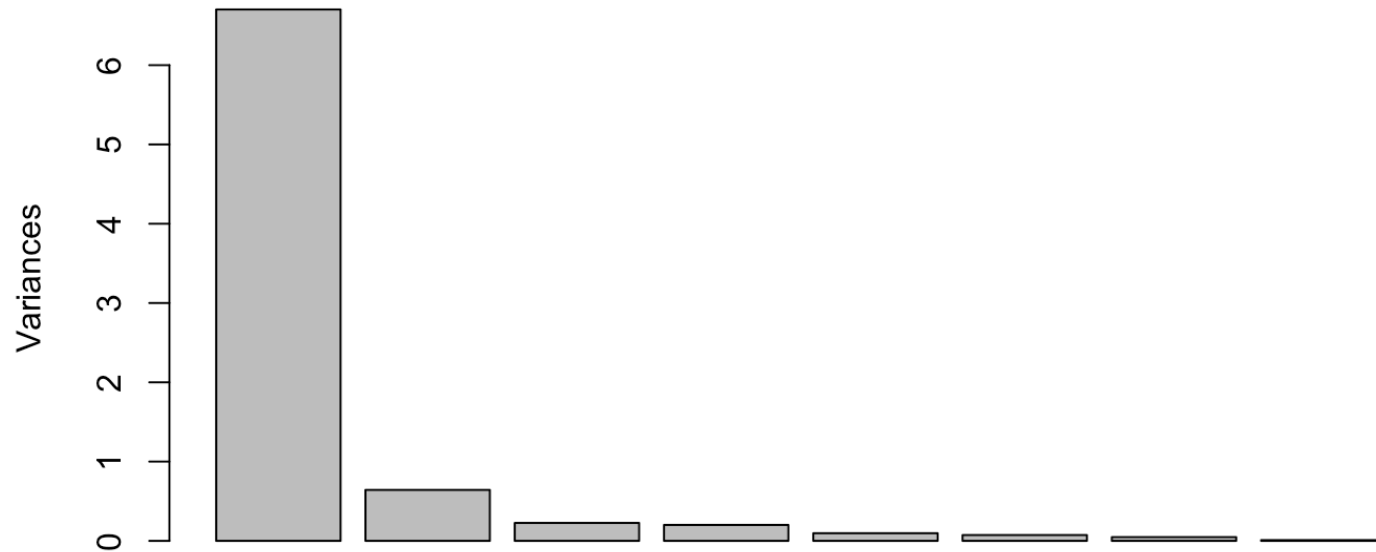

The amount of variability explained

```
aa <- summary(pca3)
round(aa$importance[, c(1, 2, 3, 8)], 4)
```

##	PC1	PC2	PC3	PC8
## Standard deviation	2.5890	0.8017	0.4754	0.0989
## Proportion of Variance	0.8378	0.0803	0.0283	0.0012
## Cumulative Proportion	0.8378	0.9182	0.9464	1.0000

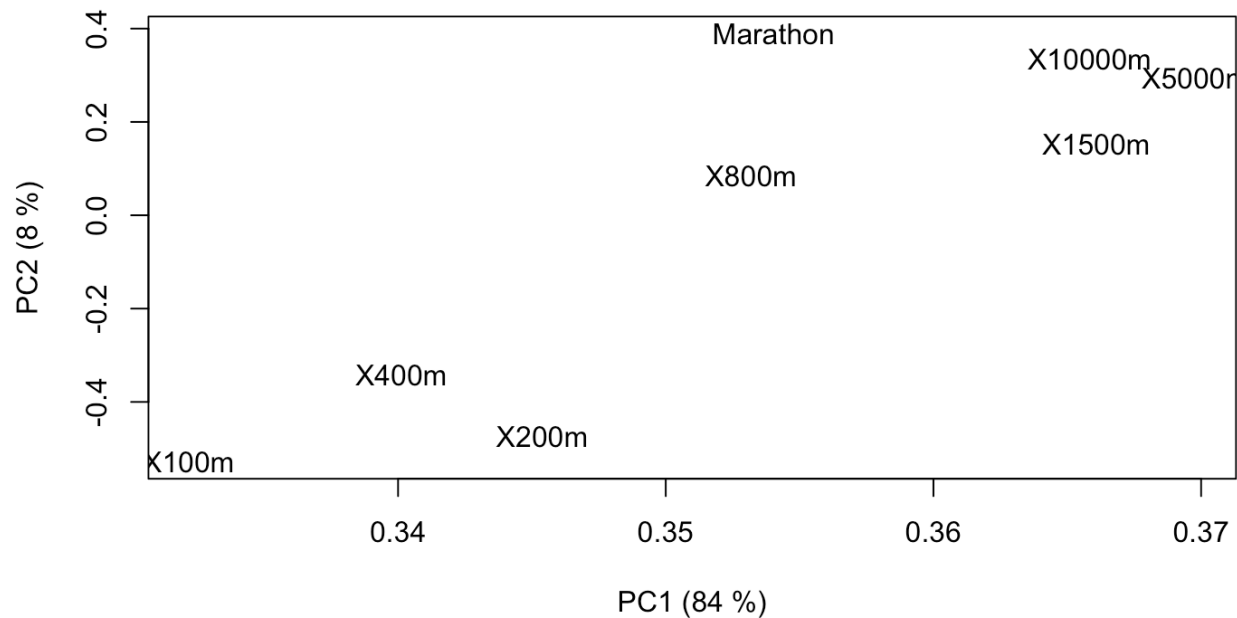
Scree plot

```
plot(pca3, main = " ")
```



Loading plot

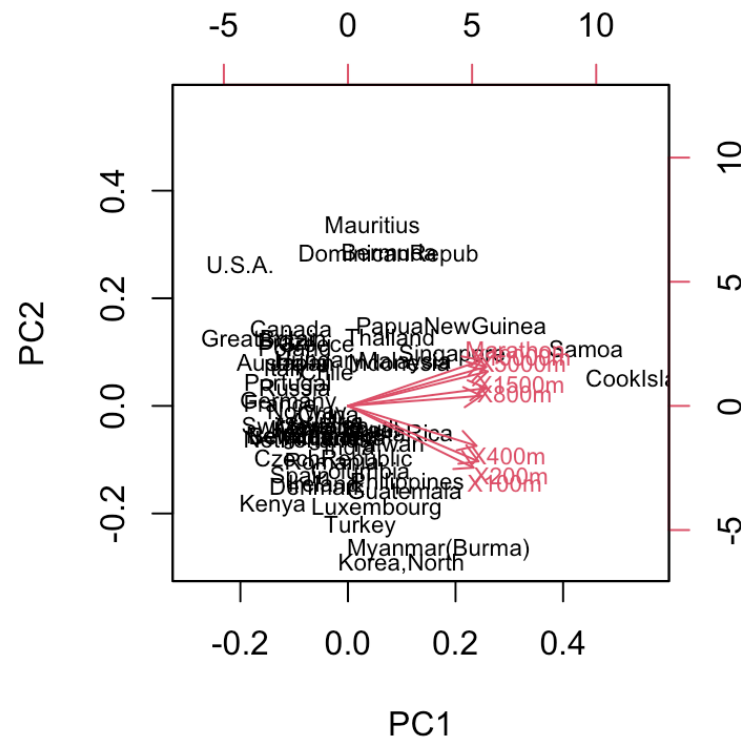
```
loadingplot(pca3, main = "", comps = c(1,2),  
            scatter = TRUE, labels = 'names')
```



This is more a picture of the correlation structure of the variables.

Biplot

```
biplot(pca3, choices = c(1,2), main = "''", cex = 0.7)
```



PC1 and PC2 explain most (92%) of the variability: may be interpreted

```
pca3$rotation[,1:2]
```

##		PC1	PC2
##	X100m	0.3321807	-0.52791926
##	X200m	0.3453840	-0.47375720
##	X400m	0.3401182	-0.34108719
##	X800m	0.3531835	0.08588801
##	X1500m	0.3660851	0.15199712
##	X5000m	0.3697992	0.29437373
##	X10000m	0.3658366	0.33445336
##	Marathon	0.3540349	0.38925581

- PC1: Weighted average, ranking of countries.
- PC2: Distinguishes short from long distances.

Ranking countries based on PC1

```
PC1 <- pca3$x[,1]
X100m.Seconds <- 60*TrackRecords$X100m
records <- data.frame(TrackRecords,
                      PC1 = PC1, X100m.Seconds)
ord <- order(PC1)
head(records[ord, c("PC1", "Marathon", "X100m.Seconds")])
```

##	PC1	Marathon	X100m.Seconds
## U.S.A.	-3.811745	125.38	9.78
## GreatBritain	-2.967663	127.13	9.84
## Kenya	-2.660737	124.55	10.26
## France	-2.395852	126.36	10.02
## Australia	-2.307853	127.51	9.96
## Italy	-2.213433	127.29	10.02